CY4001: Industrial Chemistry

Unit 3:Fuel Chemistry

Combustion

Pierre Louis Dulong

- Pierre Louis Dulong was a French chemist
- Orphaned at age 4, yet grew up with strong determination to learn
- Overcame financial struggles that forced him to quit medical studies
- Redirected his passion into chemistry and made major scientific contributions
- Discovered key principles on heat capacity, gas expansion, and calorimetry
- Remembered today as one of the 72 scientists honored on the Eiffel Tower, proving that perseverance can lead to greatness

1785 - 1838

At the time of his death, he was working on the development of precise methods in calorimetry. His last paper, published the year of his death, examined the heat released from chemical reactions

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Combustion

- "Combustion is the sequence of exothermic chemical reactions between a fuel and an oxidant accompanied by the production of heat and conversion of chemical species"
- Combustion is a chemical process accompanied by the liberation of heat and light. It is necessarily an exothermic reaction.

$$C + O_2 \longrightarrow CO_2 + 94 \text{ kcal}$$

$$2H_2 + O_2 \longrightarrow 2H_2O + 68.5 \text{ kcal}$$

The calorific value of a fuel depends mainly on the amount of carbon and hydrogen.

Difference between burning and combustion

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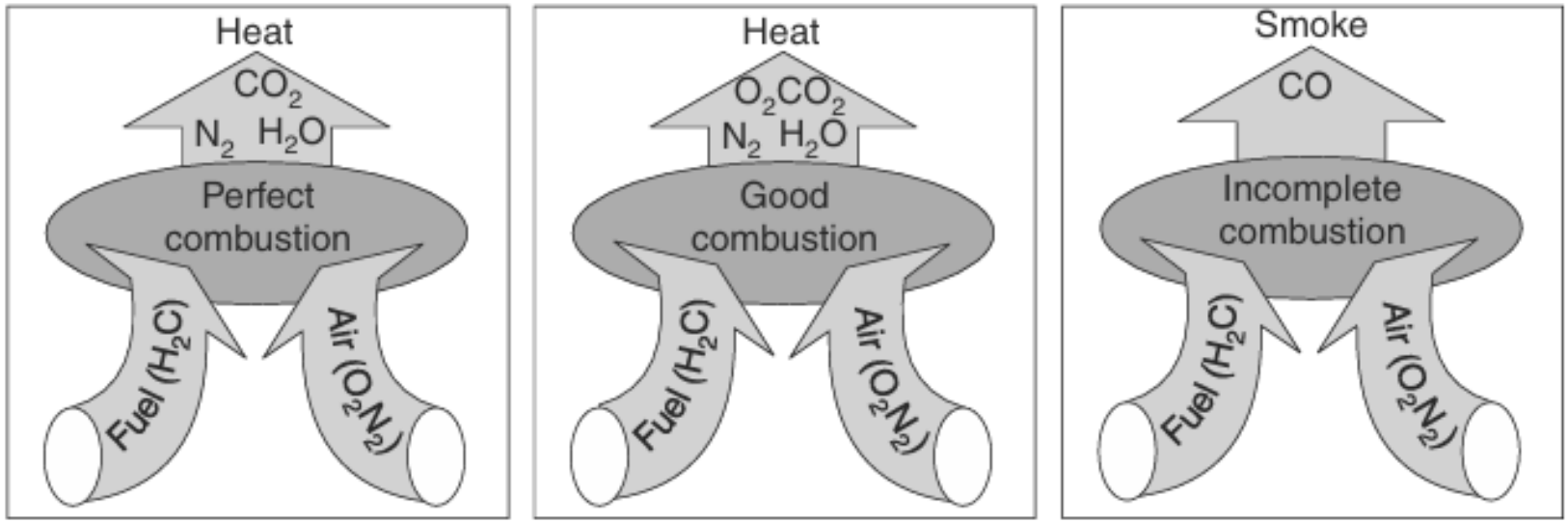
Principle of Combustion

- Combustion refers to the rapid oxidation of fuel accompanied by the production of heat, or heat and light. Complete combustion of a fuel is possible only in the presence of an adequate supply of oxygen.

$$C + O_2 \longrightarrow CO_2 + 8080 \text{ kcal/kg of carbon}$$
$$2C + O_2 \longrightarrow 2\,CO + 2430 \text{ kcal/kg of carbon}$$
$$2H_2 + O_2 \longrightarrow 2H_2O + 34500 \text{ kcal/kg of hydrogen}$$
$$S + O_2 \longrightarrow SO_2 + 2240 \text{ kcal/kg of sulphur}$$

Types of combustion



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Calorific Value

The calorific value of a fuel is defined as "the quantity of heat (expressed in *cal*) liberated by the complete combustion of the unit weight (1 gm) of the fuel in air or oxygen".

Calorie: - the amount of heat required to raise the temperature of 1 g of water by 1°C (*i.e.* 15 to 16°C).

The calorific value of fuel depends upon the nature of the fuel and the relative proportions of the elements present.

(i) Increases as the amount of hydrogen in the fuel increases

(ii) Moisture if present considerably reduces the calorific value of a fuel.

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Calorific Value

(a) Gross Calorific Value (GCV) or High Calorific Value (HCV)

- Gross Calorific Value is the total amount of heat released when 1 kg of fuel is completely burned.
- It includes the latent heat of condensation of water vapour formed during combustion. It is measured using a Bomb Calorimeter.

$$\text{Fuel} \xrightarrow[\text{Combustion, liberates heat}]{O_2} \underset{\text{Steam}}{CO_2 + H_2O\,(g)} \underset{\text{Heat 1}}{+\,\Delta} \xrightarrow[\text{Steam condensed to water and liberates heat}]{\text{Cooling to RT}} H_2O + \underset{\text{Heat 2}}{\text{latent heat}}$$

GCV assumes that the water produced during burning cools and condenses, releasing additional heat.

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Calorific Value

(a) Gross or High Calorific Value (GCV or HCV)

Dulong's Formula for the Calculation of Calorific Value

Calorific value of a fuel is the sum of the calorific values of all the combustible constituents present in the fuel.

$$\text{GCV or HCV} = \frac{1}{100}\left[8080\,C + 34500\left(H - \frac{O}{8}\right) + 2240\,S\right] \text{kcal/kg}$$

C, H, O, S refers to the percentage of carbon, hydrogen, oxygen and sulphur, respectively.

- It is based on the assumption that the calorific values of C, H and S are found to be 8080, 34500 and 2240 kcal, when 1 kg of the fuel is burnt completely.
- However, all the oxygen in the fuel is assumed to be present in combination with hydrogen in the ratio H : O as 1 : 8 by weight. So the surplus hydrogen available for combustion is H – O / 8

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Calorific Value

(b) Net or Lower Calorific Value (NCV or LCV)

- Net Calorific Value, also called Lower Calorific Value (LCV), is the actual usable heat released when 1 kg of fuel is completely burned.
- It does not include the latent heat of condensation of the water vapour formed during combustion.
- In NCV, the water produced stays in vapour form, so its heat is not recovered.

NCV = GCV – Latent heat of condensation of steam produced

$$\text{Fuel} \xrightarrow[\text{Combustion, liberates heat}]{O_2} \underbrace{CO_2 + H_2O\,(g)\uparrow}_{\text{Steam escapes}} + \underbrace{\Delta}_{\text{Heat}}$$

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Calorific Value

(b) Net or Lower Calorific Value (NCV or LCV)

Latent heat of condensation of steam

Consider the following combustion reaction:

$$\underset{\substack{2\text{ g}\\1}}{H_2} + 1/2\ \underset{\substack{16\text{ g}\\8}}{O_2} \longrightarrow \underset{\substack{18\text{ g}\\9}}{H_2O}$$

(1 kg of hydrogen produces 9 kg of water)

The latent heat of condensation of steam per kg = 587 kcal

The latent heat of condensation of steam produced by H amount of hydrogen is $= \frac{9}{100} H \times 587 \quad = 0.09\ H \times 587$

where, H = % of H_2 in the fuel.

$$\text{NCV} = \text{GCV} - \frac{9}{100}\text{H} \times 587\ \text{kcal/kg}$$

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Calorific values

Problem 1

- **Calculate the gross and net calorific values of a coal sample having the following compositions: C = 80 %, H_2 = 8 %, O_2 = 8 %, S = 2 %, and ash = 2%. Latent heat of steam is = 587 cal/g.**

(i) Gross Calorific Value (GCV)

$$GCV = \frac{1}{100}\left[8080\,C + 34500\left(H - \frac{O}{8}\right) + 2240\,S\right] \text{kcal/kg}$$

$$= \frac{1}{100}\left[8080 \times 80 + 34500\left(8 - \frac{8}{8}\right) + 2240 \times 2\right] \text{kcal/kg}$$

$$= \frac{1}{100}\,[646400 + 241500 + 4480]\ \text{kcal/kg}$$

$$= \frac{1}{100}\,[892380]\ \text{kcal/kg}$$

$$= 8923.8\ \text{kcal/kg}$$

Problem 1

(ii) Net Calorific Value (NCV)

$$= GCV - \frac{9}{100} H \times 587 \text{ kcal/kg}$$

$$= 8923.8 - \frac{9}{100} \times 8 \times 587 \text{ kcal/kg}$$

$$= 8923.8 - 422.64 \text{ kcal/kg}$$

$$= 8501.16 \text{ kcal/kg}$$

Some Problems to try

1. On analysis, a coal sample has the following composition by weight: C = 75 %, O2 = 04 %, S = 05 %, and ash = 3%. Net calorific value of the fuel is 9797.71 kcal/kg. Calculate the percentage of hydrogen and gross calorific value of coal.

2. A coal sample on analysis gives: C = 68%, S = 2%, O = 6%, N = 1%, Ash = 8%, Moisture = 5%. Its Net Calorific Value (NCV) is found to be 7030 kcal/kg. Calculate (a) The percentage of hydrogen (H) in the coal (b) the Gross Calorific Value (GCV)

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How do we measure this experimentally?

A calorimeter is used for determining calorific value. For determining calorific value of solid and liquid fuels a bomb calorimeter is used and for gaseous fuel Junker's calorimeter is used.

Calorimeter

A calorimeter is a device that measures the amount of heat involved in a chemical or physical process.

Principle

A bomb calorimeter works on the principle that the heat released by the complete combustion of fuel at constant volume is equal to the heat absorbed by the surrounding water and calorimeter, causing a measurable rise in temperature.

Heat released by fuel=Heat absorbed by water + calorimeter

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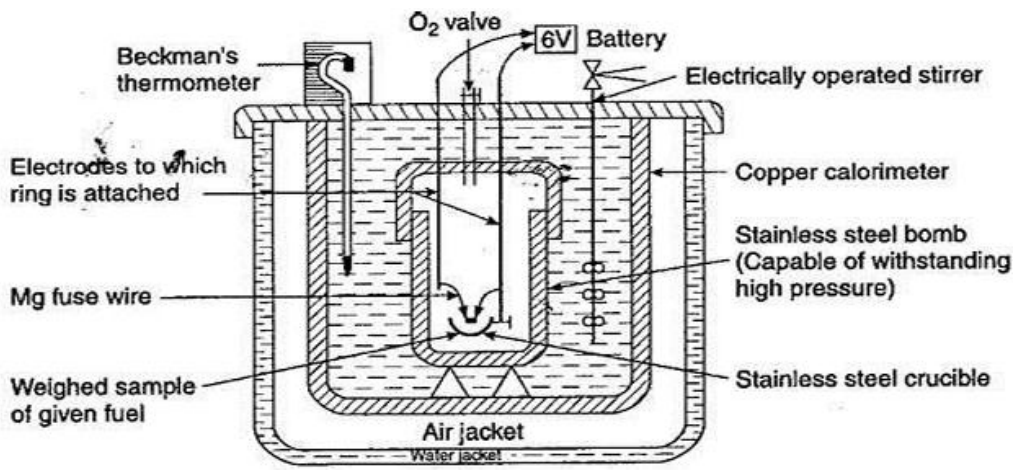
Determination of Calorific Value

Bomb Calorimeter (For solid and non-volatile liquid fuels)

- The bomb calorimeter method is a way to measure the heat released when a fuel is burned.
- It uses a pressurized oxygen chamber, called a bomb, to burn the fuel and a water bath to measure the heat released.



How it works

1. Place a fuel sample in the bomb
2. Fill the bomb with oxygen
3. Ignite the sample with a hot wire
4. Place the bomb in a water bath
5. Measure the temperature of the water
6. Record the temperature change until it stabilizes
7. Calculate the heat of combustion using the temperature change

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Determination of Calorific Value

Bomb Calorimeter (For solid and non-volatile liquid fuels)

Procedure

- Weigh a known mass of the fuel sample and place it in the crucible; connect the fuse wire so it touches the sample.
- Fill the bomb with pure oxygen (about 20–25 atm) to ensure complete combustion, then seal it.
- Immerse the bomb in a known mass of water inside the calorimeter and record the initial temperature.
- Ignite the fuel electrically; allow complete combustion while continuously stirring and note the maximum (final) temperature.
- Calculate GCV using the rise in temperature and the total heat-absorbing capacity $(W+w)$(W+w) of water and calorimeter.

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Determination of Calorific Value

Bomb Calorimeter (For solid and non-volatile liquid fuels)

GCV is calculated by measuring how much the water and calorimeter heat up when a known mass of fuel burns, and dividing that total heat by the mass of fuel used.

$$\text{GCV} = \frac{(W + w)\Delta T}{m}$$

W = mass of water in the calorimeter
w = water equivalent of calorimeter
ΔT = rise in temperaturem = mass of fuel burned
m = mass of the fuel

Water equivalent is the amount of water that would absorb the same amount of heat as the calorimeter (its metal walls, stirrer, thermometer, etc.) for the same rise in temperature.

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Determination of Calorific Value

Problems on Bomb calorimeter:

1. Calculate the gross and net calorific value of a coal sample from the following data obtained from bomb calorimeter. Weight of coal 0.73 g, weight of water in the calorimeter 1500 g, water equivalent in calorimeter 470 g, initial temp 25 ^{0}C and final temp 28^{0}C, percentage of hydrogen in coal 2.5% and latent heat of steam 587 cal/g.

Solution: The given data

m= 0.73 g, W = 1500 g, w= 470 g, t_1 = 25^{0}C t_2 =28^{0}C, % of hydrogen = 2.5%, Latent heat of steam= 587 cal/g.

Gross calorific value (GCV) = $(W+w)(t_2 - t_1) / m$ cal/g.

$= (1500+470)(28-25) / 0.73$

$= 8095.89$ cal/g.

Net calorific value (NCV) = $GCV - 0.09 \times H \times 587$

$= 8095.89 - 0.09 \times 2.5 \times 587$

$= 7963.82$ cal/g.

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Determination of Calorific Value using Bomb Calorimeter

O_2 Bomb Calorimeter
at IIT PKD

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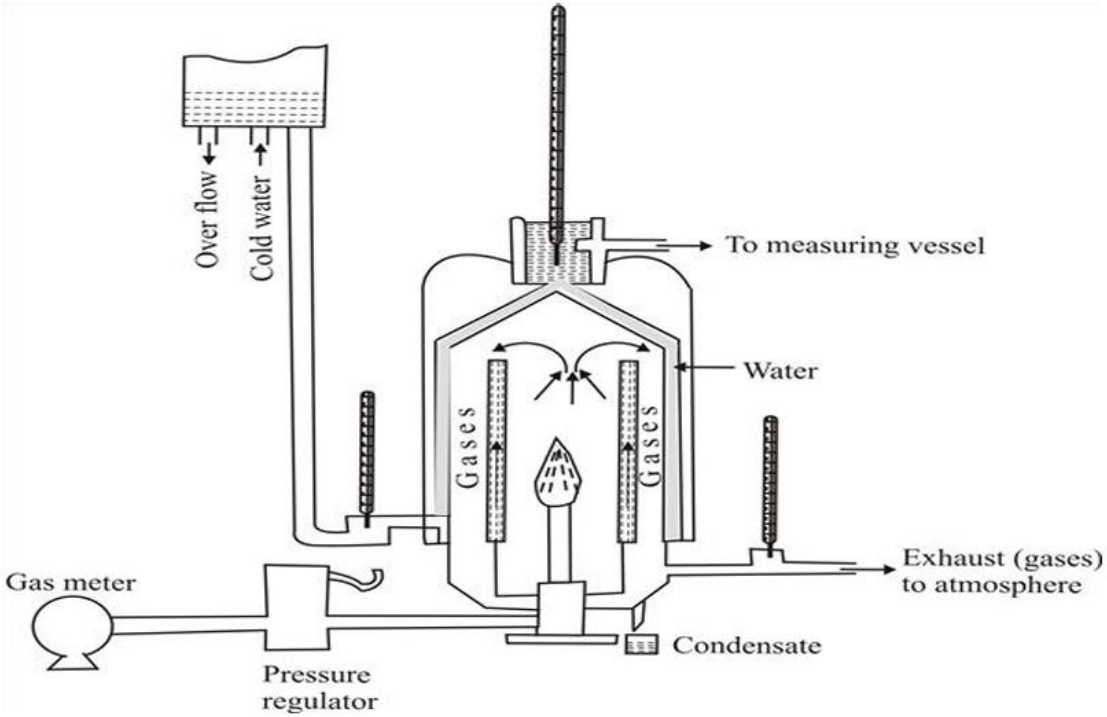
Determination of Calorific Value

Junkers Gas Calorimeter (For gaseous and volatile liquid fuels)

- A known volume of gas is burned
- The heat is transferred to a known volume of water that's flowing steadily
- The rise in temperature of the water is measured
- The calorific value of the gas is calculated using a formula



A calorimeter with a burner on a tripod stand
A gas flow meter
A pressure governor
A digital temperature indicator
A measuring jar
Rubber tubing for connecting the gas and water

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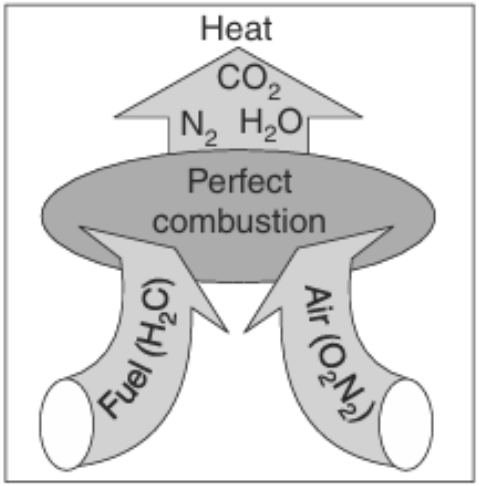
Combustion efficiency



Combustion efficiency is the percentage of a fuel's heat content that is converted into usable heat energy. It's a measure of how well a fuel is burned and how much harmful pollution is produced.

Some factors that affect combustion efficiency

- **Fuel type:** Amount of hydrogen, carbon, and oxygen
- **Excess air:** Lowers the flame temperature
- **Lack of air:** Incomplete combustion
- **Stack temperature:** The temperature of the gases that exit

Stoichiometric air-fuel ratio

The stoichiometric air-fuel ratio is the ideal ratio of air to fuel for complete combustion in a thermal engine.

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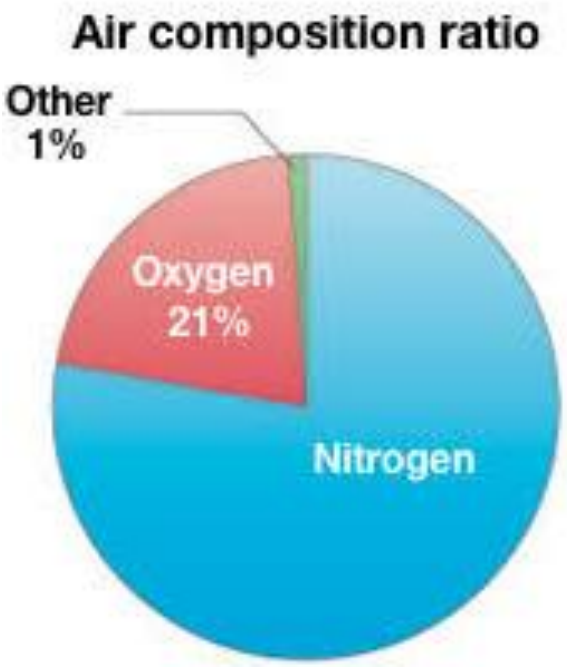
How much air is too much air?

Theoretical Calculation of Minimum Air Required for Combustion

Air contains 21% of oxygen by volume and 23% of oxygen by weight. Hence, from the amount of oxygen required by the fuel, the amount of air can be calculated.

Amount of oxygen required

- The amount of oxygen required for complete combustion of a fuel can be calculated from the combustion equations.
- By calculating the weight and volume of oxygen required for complete combustion of a fuel, it is easy to calculate the volume and weight of air required for the complete combustion.



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Theoretical Calculation of Minimum Air Required for Combustion

Combustion of Carbon

$$C + O_2 \longrightarrow CO_2$$

By volume	1 vol	1 vol	1 vol
By weight	12 g	32 g	44 g

1 part by volume of carbon requires 1 part by volume of oxygen for complete combustion.

(or)

12 g of carbon requires 32 g of oxygen for complete combustion (to give 44 g of CO_2).

1 g of carbon requires 32/12 g of oxygen for complete combustion.

$\therefore$ C g of carbon requires $= \frac{32C}{12}$ g of oxygen

(or)

C parts by weight of carbon requires $= \frac{32C}{12}$ parts by weight of O_2

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Theoretical Calculation of Minimum Air Required for Combustion

Combustion of Hydrogen

	H_2	+ $1/2\ O_2$	$\longrightarrow H_2O$
By volume	1 vol	0.5 vol	1 vol
By weight	2 g	16 g	18 g

1 part by volume of hydrogen requires 0.5 part by volume of oxygen for complete combustion.

(or)

4 g of hydrogen requires 32 g of oxygen for complete combustion.

1 g of hydrogen requires 32/4 g of oxygen for complete combustion.

$\therefore$ **H** g of hydrogen requires $= \frac{32 \times H}{4}$ g of O_2

(or)

H parts by weight of hydrogen requires $= \frac{32 \times H}{4}$ parts by weight of

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Theoretical Calculation of Minimum Air Required for Combustion

Role of Oxygen in Combustion of Hydrogen

- The oxygen present in the fuel consumes some amount of hydrogen to form water. So, the quantity of hydrogen taken up by oxygen will not take part in the combustion reaction. Therefore, the quantity of hydrogen taken up by the oxygen will be deduced from the total hydrogen in the fuel.
- Now, the quantity of H available for the combustion reaction will be H – O/8 where H is the total quantity of hydrogen and O is the total quantity of oxygen in the fuel. (in water, the quantity of hydrogen in combination with oxygen is one-eighth of the weight of oxygen).

$$\therefore \left(H - \frac{O}{8}\right) \text{ parts by weight of hydrogen requires } \frac{H - \frac{O}{8} \times 32}{4} = 8\left(H - \frac{O}{8}\right) \text{ parts by weight of } O_2$$

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Theoretical Calculation of Minimum Air Required for Combustion

Combustion of Sulphur

$$S + O_2 \longrightarrow SO_2$$

By volume	1 vol	1 vol	1 vol
By weight	32 g	32 g	64 g

1 part by volume of sulphur requires 1 part by volume of oxygen for complete combustion.

(or)

32 g of sulphur requires 32 g of oxygen for complete combustion.

S g of sulphur requires $= \frac{32 \times S}{32} = S$ g of O_2

(or)

$\therefore$ **S** parts by weight of sulphur requires $= \frac{32 \times S}{32} = S$ parts by weight of O_2

Combustion of Nitrogen

Since nitrogen do not undergo combustion, the oxygen requirement is nil.

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Theoretical Calculation of Minimum Air Required for Combustion

Theoretical amount of oxygen required

Consequently, the theoretical amount of oxygen required for the complete combustion of 1kg of solid or liquid fuel is

$$\text{Theoretical minimum } O_2 = \left[\frac{32}{12} \times C + 8\left(H - \frac{O}{8}\right) + S\right] \text{kg}$$

Since mass % of O_2 in air is 23, the amount of air required theoretically for combustion of 1 kg of the fuel is,

$$\text{Air (theoretical) (by weitht)} = \frac{100}{23}\left[\frac{32}{12} \times C + 8\left(H - \frac{O}{8}\right) + S\right] \text{kg}$$

$$\text{Air (theoretical) (by volume)} = \frac{100}{21}\left[\frac{32}{12} \times C + 8\left(H - \frac{O}{8}\right) + S\right] \text{litre}$$

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Combustion

Problem-1

Calculate the mass of air needed for complete combustion of 5 kg of coal that contains C = 80%, H = 15%, O = rest.

$$\text{Air (theoretical) (by weight} = \frac{100}{23}\left[\frac{32}{12} \times C + 8\left(H - \frac{O}{8}\right) + S\right] \text{kg}$$

Answer: 71.37 kg

Combustion

Problem-2

Calculate the volume of air required for complete combustion of 1 m^3 of a gaseous fuel having the composition: CO = 46%, CH_4 = 10%; H_2 = 40%, C_2H_2 = 2%, N_2 = 1% and the remaining being CO_2.

Ans.: Let the gaseous fuel be 1 m^3

$$CO + \frac{1}{2}O_2 = CO_2$$

$$\text{Volume of } O_2 \text{ required} = 0.46 \times \frac{1}{2} = 0.23\,m^3$$

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_20$$

$$\text{Volume of } O_2 \text{ required} = 0.1 \times 2 = 0.2\ m^3$$

$$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$$

$$\text{Volume of } O_2 \text{ required} = 0.4 \times \frac{1}{2} = 0.2\,m^3$$

$$C_2H_2 + 2\left(\frac{1}{2}O_2\right) \rightarrow 2CO_2 + H_20$$

$$\text{Volume of } O_2 \text{ required} = 0.2 \times \frac{5}{2} = 0.05\,m^3$$

$$\text{Total } O_2 \text{ required} = (0.23 + 0.2 + 0.05)\ m^3$$

$$= 0.68\ m^3$$

$$\text{Volume of air required} = 0.68 \times \frac{100}{21}\,m^3$$

$$= 3.238\ m^3$$

Combustion

Problem-3

Calculate the weight and volume of air required for the combustion of 2 kg of carbon.

Ans.: Combustion reaction: $\underset{12g}{C} + \underset{32g}{O_2} \rightarrow CO_2$

Weight of oxygen required to burn 12 kg of C = 32 kg.

Weight of oxygen required to burn 1 kg of C = 2.67 kg.

$$\text{Weight of air required} = 2.66 \times \frac{100}{23}$$

$$= 11.60 \text{ kg} = 11600 \text{ g}$$

Volume of air required for combustion of 2 kg of C

$$= \frac{11600}{\text{molar mass of air}} \times 22.4\text{L} \quad = \frac{11600}{28.94} \times 22.4\text{L} = 8.97\,\text{m}^3$$

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Flue-gas Analysis

- During combustion, a fuel is burnt in the combustion chamber with the help of air from the atmosphere. The mixture of gases (like CO_2, O_2, CO, etc.) coming out from the combustion chamber is called flue gas.

- Flue gas analysis is a process that measures the composition of gases in a flue to evaluate the efficiency of a combustion process. It's used to optimize the ratio of air to fuel, which can help reduce fuel consumption and emissions.

Optimizing fuel efficiency: Flue gas analysis helps to ensure the right balance of air and fuel for efficient combustion

Reducing emissions: Flue gas analysis helps to ensure that emissions meet environmental standards

Monitoring safety: Flue gas analysis helps to detect incomplete combustion and unburnt fuels

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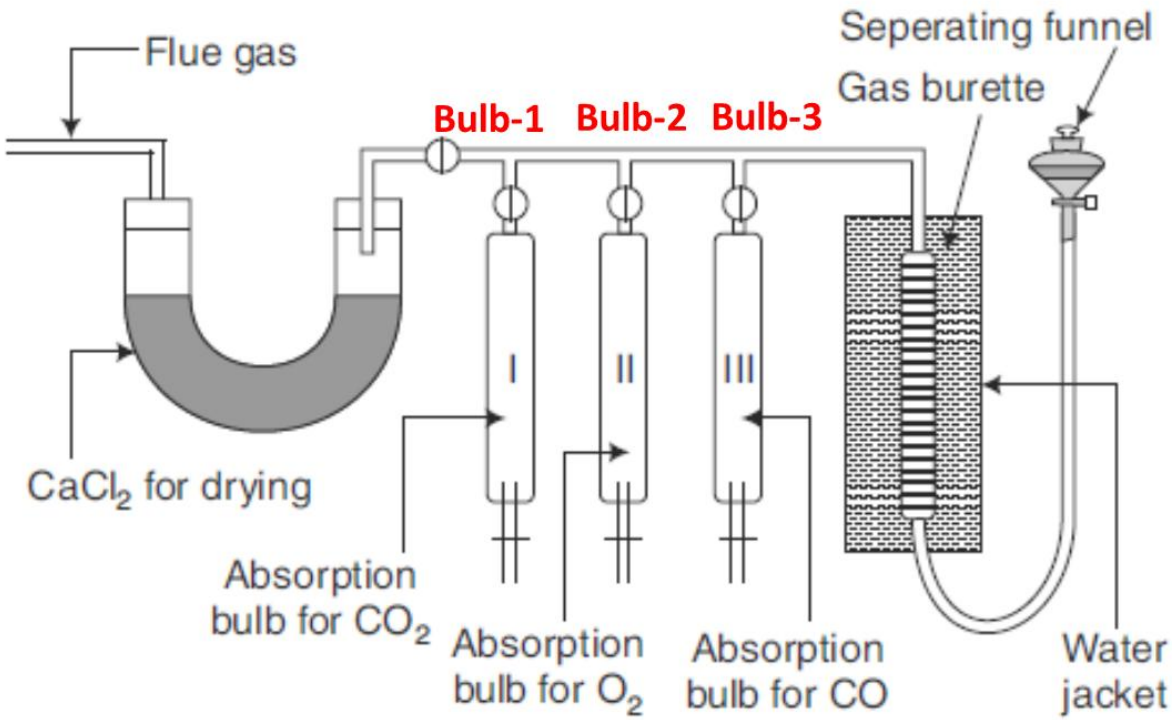
Flue-gas Analysis [Orsat's Method]

Orsat's apparatus is used for the analysis of flue gas.



- **Bulb-1:** Consists of 'potassium hydroxide' solution and it absorbs CO_2.
- **Bulb-2:** Consists of 'alkaline pyrogallol' solution and it absorbs O_2.
- **Bulb-3:** consists of 'ammoniacal cuprous chloride' solution and it absorbs CO.

The remaining gas in the burette after the absorption of CO_2, O_2, and CO is taken as nitrogen.

Significance of Flue-gas Analysis

- Based on the composition of the flue gas, it is possible to predict the complete or incomplete combustion process.
- If the flue gas contains any CO, it indicates the incomplete combustion in the combustion chamber and short supply of oxygen.
- If the flue gas contains O_2, it indicates excess supply of air and it is essential to reduce the air supply.

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Important Terms Related to Combustion

Ignition Temperature:

It is defined as the lowest temperature to which the fuel must be heated, so that it starts burning smoothly.

- Besides sufficient supply of oxygen or air, every fuel requires heating to a definite temperature to start active combustion. This is known as ignition temperature.
- The ignition temperature of coal is about 300°C.
- In the case of liquid fuels, the ignition temperature is called the flash point, which ranges from 200 – 400°C.
- For gaseous fuels, the ignition temperature is in the order of 800°C.

Unit 3: Fuel Chemistry

Important Terms Related to Combustion

Spontaneous Ignition Temperature (SIT)

It is defined as the minimum temperature at which the fuel catches fire (ignites) spontaneously without external heating.

- If the heat evolved in a system is unable to escape, temperature of the system goes on increasing and when SIT is reached, the system burns on its own.
- Fuels which have low ignition temperature can catch fire very quickly. On the other hand, if the ignition temperature is high, it is difficult to ignite the fuel.

Example: If coal is stored in such a way that heat is slowly dissipated, slow oxidation may produce sufficient heat to raise the temperature to ignite on its own.

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Important Terms Related to Combustion

Explosive Range or Limits of Inflammability

Most of the gaseous fuels have two percentage limits, called upper limit and lower limit. Those limits represent the percentage by volume of fuel present in fuel–air mixture. The range covered by these limits is termed the explosive range of the fuel.

Calorific Intensity or Flame Temperature

It is the maximum temperature reached when the fuel is completely burnt in the theoretical amount of air.

When the fuel burns with a small flame, the entire heat is concentrated in a small area and the calorific intensity will be high. But while burning solid and liquid fuels as well as some gaseous fuels, much volatile matter is given off resulting in a large flame. Here the heat liberated gets distributed over a wide area and hence the calorific intensity will be low.